

Lecture 4: Vector-Valued Functions

Background & Motivation

We now shift from static geometry to motion. A vector-valued function $\mathbf{r}(t)$ traces a curve through space as the parameter t varies—think of it as the position of a particle at time t . This chapter connects the vector algebra from Chapter 1 to calculus: we'll differentiate and integrate these functions component by component, leading to tangent vectors, arc length, curvature, and projectile motion.

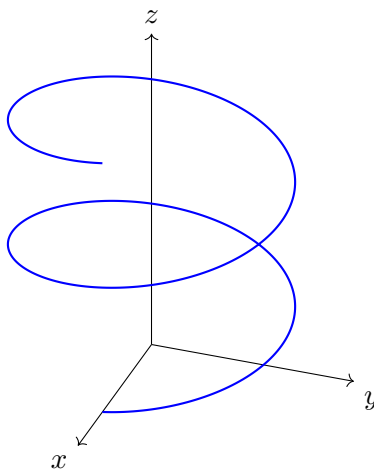
1. Parametric Curves in $\mathbb{R}^3$

Definition 1: Vector-Valued Function (VVF)

A VVF maps a scalar t to a vector:

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\hat{\mathbf{i}} + g(t)\hat{\mathbf{j}} + h(t)\hat{\mathbf{k}}$$

- f, g, h are the _____
- As t varies, the tip of $\mathbf{r}(t)$ traces a **space curve**



Helix $\mathbf{r}(t) = \langle R \cos t, R \sin t, ct \rangle$ spiraling around the z -axis. As t increases, $\mathbf{r}(t)$ rotates in the xy -plane while rising linearly in z .

Common Parametric Curves

- **Line:** $\mathbf{r}(t) = \mathbf{r}_0 + t \mathbf{d}$
- **Circle** (radius R , xy -plane): $\mathbf{r}(t) = \underline{\hspace{2cm}}$, $t \in [0, 2\pi]$
- **Helix** (radius R , rising rate c): $\mathbf{r}(t) = \underline{\hspace{2cm}}$
- **Ellipse:** $\mathbf{r}(t) = \langle a \cos t, b \sin t, 0 \rangle$

Example 1: Identifying a Helix

Describe the curve $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, 2t \rangle$ for $t \geq 0$.

2. Derivatives of VVFs

Definition 2: Derivative of a VVF

The derivative $\mathbf{r}'(t)$ is the instantaneous rate of change of the position vector, obtained by differentiating each component separately. Geometrically it is the **velocity vector**: it points in the direction the curve is moving at that instant, and its magnitude is the speed at which the particle is traversing the curve.

$$\mathbf{r}'(t) = \underline{\hspace{2cm}}$$

- $\mathbf{r}'(t)$ is a $\underline{\hspace{2cm}}$ to the curve at $\mathbf{r}(t)$
- $\|\mathbf{r}'(t)\|$ is the $\underline{\hspace{2cm}}$

Derivative Rules for VVFs

- Scalar product rule: $\frac{d}{dt}[f(t) \mathbf{r}(t)] = \underline{\hspace{2cm}}$
- Dot product rule: $\frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{s}(t)] = \underline{\hspace{2cm}}$
- Cross product rule: $\frac{d}{dt}[\mathbf{r}(t) \times \mathbf{s}(t)] = \underline{\hspace{2cm}}$
- Chain rule: $\frac{d}{dt}[\mathbf{r}(g(t))] = \mathbf{r}'(g(t)) \cdot g'(t)$

Example 2: Derivative and Analysis at a Point

Let $\mathbf{r}(t) = \langle t^3, \sin t, e^t \rangle$. Find $\mathbf{r}'(t)$ and analyze at $t = \pi$.

3. Integrals of VVFs

Definition 3: Integral of a VVF

Also **component-wise**:

$$\int \mathbf{r}(t) dt = \underline{\hspace{2cm}}$$

– $\mathbf{C} = \langle C_1, C_2, C_3 \rangle$ is a **vector constant** of integration

$$\int_a^b \mathbf{r}(t) dt = \underline{\hspace{2cm}}$$

Example 3: Definite Integral of a VVF

Evaluate $\int_0^1 \langle 3t^2, 2 \sin t, e^{2t} \rangle dt$.

4. Tangent Lines and Curves on Surfaces

Tangent Line at $t = t_0$

$$\mathbf{L}(s) = \underline{\hspace{2cm}}$$

– Point of tangency: $\mathbf{r}(t_0)$ – Direction: $\mathbf{r}'(t_0)$

– To show a curve lies on a surface: substitute the components into the surface equation and verify

Example 4: Tangent Line + Curve on a Surface

The curve $\mathbf{r}(t) = \langle e^t, e^{-t}, t \rangle$. Show it lies on $xy = 1$ and find the tangent line at $t = 0$.

Show all work for full credit.

Practice Problems

Problem 1. If $\mathbf{r}(t) = \langle t^2, \ln(t+1), \cos t \rangle$, find $\mathbf{r}'(t)$ and $\mathbf{r}'(0)$.

Problem 2. Evaluate $\int_0^\pi \langle \cos t, \sin t, 1 \rangle dt$.

Problem 3. A particle has velocity $\mathbf{v}(t) = \langle 2t, 3, -1 \rangle$ and $\mathbf{r}(0) = \langle 1, 0, 5 \rangle$. Find $\mathbf{r}(t)$.

Problem 4. Show that $\mathbf{r}(t) = \langle 2\cos t, 3\sin t, t \rangle$ lies on the elliptic cylinder $\frac{x^2}{4} + \frac{y^2}{9} = 1$. Find the tangent line at $t = \pi/2$.

Problem 5. Prove: if $\|\mathbf{r}(t)\| = c$ (constant), then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$ for all t .

Hint: differentiate $\mathbf{r}(t) \cdot \mathbf{r}(t) = c^2$.